

Mechanoreception of Soft Robots via Multi-channel Optical Sensing Using Ambient Light and Internal Infrared Light

Benkang Lou, Yifan Xuan, Lin Wang, Zhongfeng Qian, Rongwei Wen, Sicong Liu, Juan Yi*,
Jian S. Dai* and Zheng Wang

Abstract—Soft robots exhibit high deformability, enabling compliant interactions between the robots and unstructured environments. Mechanoreception of the robot state and its surroundings is essential for feedback control and decision making in these interactive scenarios, while remaining in significant challenges owing to the complexity of deformation and environment. In this paper, we introduce a multi-channel optical sensing approach using ambient light and internal infrared light to detect the contact force and contact position of soft robots. We demonstrate that it is possible to simultaneously measure multimodal physical quantities (external contact position, and interactive forces) with a concise optical waveguides design. This concept is verified in a soft multi-channel optical sensor design that embeds optical sensing units for receiving both environmental visible light and infrared light. The entire methodology from the core sensing principle to conceptual design, modeling, and experiments is developed and demonstrated in a soft robotic gripper capable of providing interaction sensory information, such as object detection, bending angles, and force measurements. This approach may inspire new soft sensor and robot designs to obtain multimodal sensory information for robotic applications.

I. INTRODUCTION

Soft robotics is an emerging field that has garnered considerable attention in recent years, driven by a paradigm shift from traditional rigid structures to pliable and deformable materials. The use of soft materials endows soft robots with continuous deformation capabilities, inherently enhancing their flexibility and adaptability when interacting with complex environments [1]-[6]. This continuous deformation has also inspired extensive research on soft robotic sensing to acquire rich information about the robot's state and surrounding environment, enabling the completion of various operations and tasks. In recent years, many researchers have explored various types of soft sensors using different sensing principles, such as capacitive sensors, resistive sensors, and optical waveguide sensors [7]-[13], laying the groundwork for advanced soft sensing techniques. Among these, optical waveguide sensors have demonstrated the ability to sense deformation characteristics, along with exhibiting high repeatability and low hysteresis [14]-[16], playing an

important role in soft robot interactions with the external environment.

A single soft optical waveguide sensor can measure the continuous deformation of a soft robot by outputting a voltage signal, without the need for complex calculations, and has relatively low power consumption. To obtain multimodal information from the unique deformations of soft robots, pioneering efforts have been made to incorporate several optical waveguides [17]-[19] and integrate multiple sensing mechanisms [20], demonstrating superior performance in measuring complex deformations. These efforts emphasize the use of redundant sensory inputs, aided by learning-based techniques, to characterize continuous motions and forces. However, light itself also offers a wealth of sensory inputs, such as colors and phases. Researchers have explored the possibility of utilizing these light sensory inputs to measure the bending joint positions of soft gloves and acquire additional sensory information [21]. These studies typically rely on light sources provided by internal hardware, shielding against interference from ambient light. Nevertheless, when an external object approaches the sensor without making contact ambient light can also supply valuable sensory inputs for optical waveguide sensors. When external objects interact with the sensor, some ambient light is blocked from entering, causing the intensity of ambient light propagating within the sensor to attenuate. By detecting changes in ambient light intensity, interaction information between the sensor and the external environment can be captured.

In this work, we propose a Multi-channel Soft Optical Waveguide (M-SOW) design that combines external environmental light and internally provided infrared light for detecting the contact force and contact position of soft robots. The visible light wavelength sensor, which uses environmental light provided by the external environment as input, can detect the contact position between external objects and the sensor. The infrared wavelength sensor, which uses an internal infrared emitter as input, can detect the magnitude of the external contact force. This paper presents the concept, fabrication, experimental validation of sensing performance, and a demonstration of an object being grasped by a soft robotic gripper integrated with the proposed design.

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Benkang Lou, Yifan Xuan, Sicong Liu, Juan Yi, Jian S Dai and Zheng Wang are with the Department of Mechanical and Energy Engineering, Southern University of Science and Technology, Shenzhen, China (email: 12232293@mail.sustech.edu.cn).

Juan Yi and Jian S. Dai are the corresponding authors of this work. (Juan Yi, phone: +86-13928426302, e-mail: yijuan2603@outlook.com; Jian S Dai, e-mail: daijs@sustech.edu.cn).

Juan Yi, Lin Wang, Zhongfeng Qian, and Zheng Wang are with Wisson Technology (Shenzhen) Limited Company (wanglin@wissonrobotics.com, qianzf@wissonrobotics.com)

Rongwei Wen is with the Department of Mechanical Engineering, The University of Hong Kong, Hong Kong, China (email: rwwen@hku.hk)

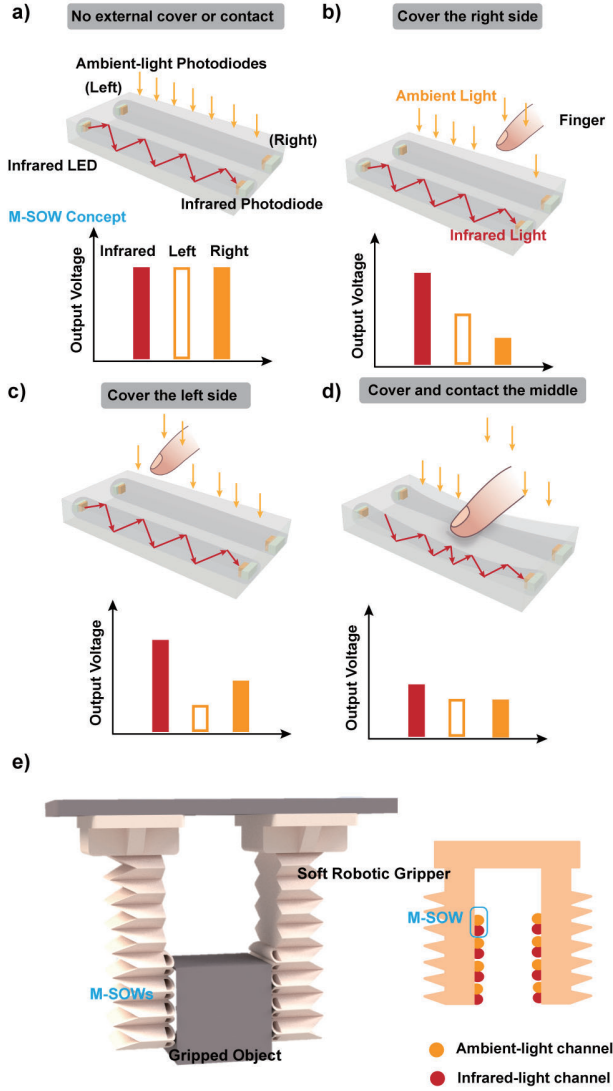


Fig. 1. Working principles and concept of sensor. (a) Concept of Multi-channel Soft Optical Waveguide (M-SOW) and sensor output voltage in the initial state. (b) -(d) Sensor output voltage response patterns for different conditions: finger covering the upper right, upper left, and contacting the center of the sensor. (e) Soft robotic gripper equipped with M-SOW and sensor configuration.

The main contributions of this work include 1) Concept of M-SOW that utilizes ambient light and internal infrared light as redundant sensory inputs for contact force and contact position measurements in soft robots; 2) the M-SOW sensor design that combines external lights and internal infrared lights, featuring a dual-light-channel waveguide with an electro-optic elements configuration consisting of a single infrared Light-Emitting Diode (LED), an infrared photodiode and two ambient-light photodiodes; and 3) the development of a soft robotic gripper with sensing capabilities to detect object contact positions and grasping forces, validating the proposed approach through a comprehensive methodology encompassing concept, design, fabrication, and system development.

This paper is structured as follows, Section II introduces the proposed conceptual design and fabrication. In Section III, we present the verification of sensing performance, including

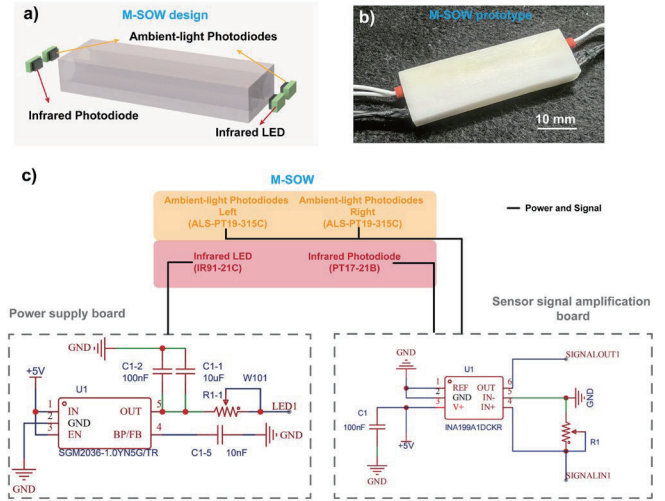


Fig. 2. Schematic of sensor design (a) Design of Multi-channel Soft Optical Waveguide (M-SOW). (b) Prototype of M-SOW. (c) Circuit design of M-SOW.

the recognition of contact position, force, and mixed contact interaction. Finally, Section IV demonstrates the verification of a soft robotic gripper, highlighting the sensing capabilities to detect object contact positions and grasping forces.

II. CONCEPTUAL DESIGN AND FABRICATION

A. Conceptual design

The M-SOW design features a dual-light-channel waveguide, as illustrated in Fig. 1(a). One channel is equipped with two photodiodes on both ends, sensitive to wavelengths from 380nm to 780nm, enabling contact position measurements along the entire channel due to local blocking of ambient light. It is designed to transmit and propagate external environmental light. This core's inner surface remains uncoated, allowing environmental light to enter the core through the sensor's outer layer. The visible light in this light path is provided by the external environment, and two identical environmental light detectors are placed at both ends of the core, without the need for additional environmental light emitters. For the ambient light sensor to work more stably, the sensor needs to be in relatively uniform and stable external ambient light conditions. When an external object encounters the sensor, the environmental light entering the channel is partially blocked, causing a change in the detected light intensity. By measuring this change, the position of the contact between the external object and the sensor can be determined, as diagramed in Fig. 1(b) and Fig. 1(c). The other channel includes an infrared LED and a photodiode sensitive to wavelengths from 780nm to 1100nm, facilitating contact force measurements because of internal light path variations caused by the soft material deformation. When the sensor is subjected to external pressure, the infrared light path within the core is compressed, leading to increased propagation loss. By measuring the changes in the signal received by the infrared photodiode, the magnitude of the external pressure can be determined, as shown in Fig. 1(d). In this way, the soft optical waveguide sensor can effectively detect both contact force and

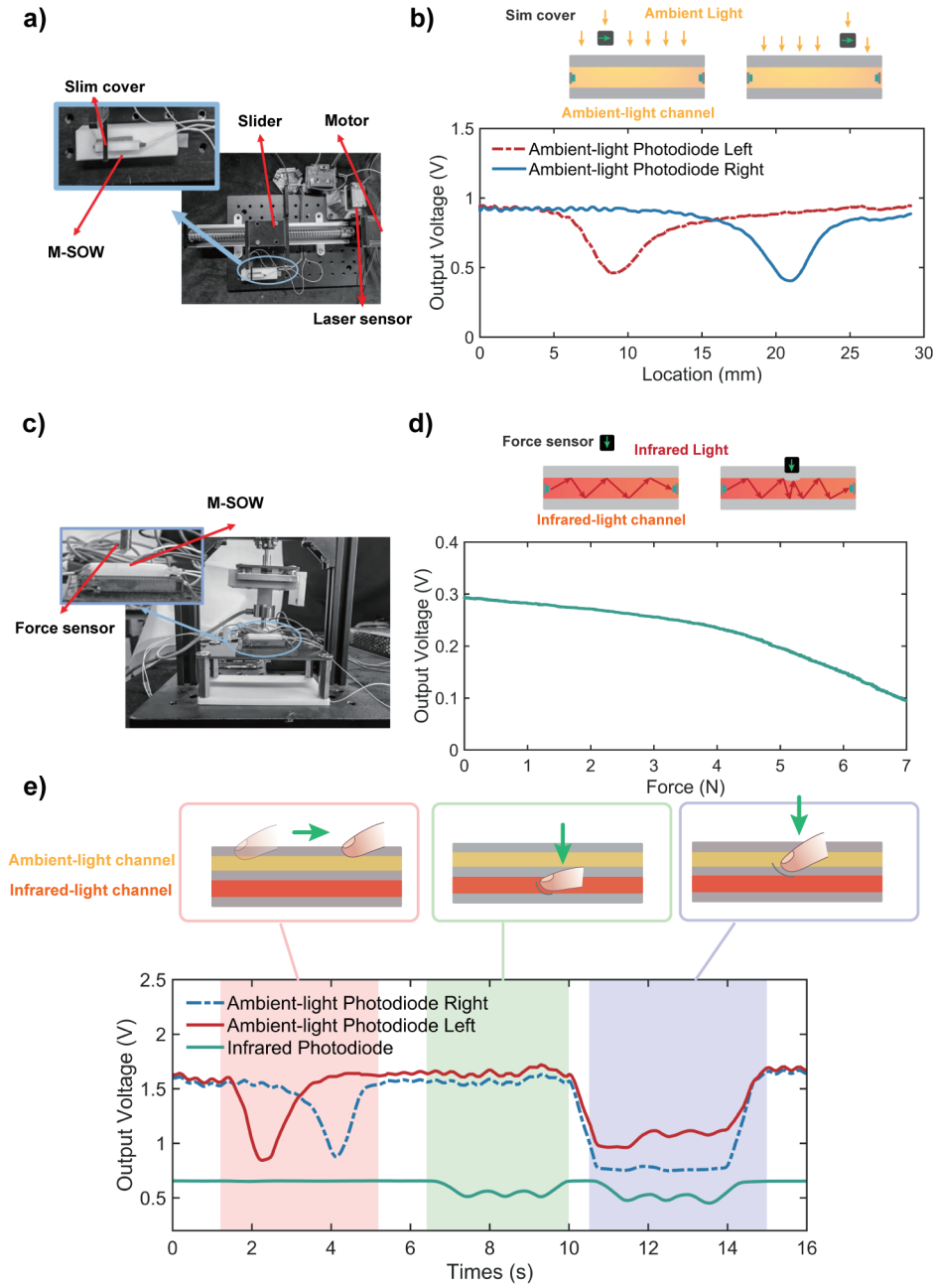


Fig. 3. Experimental setup and results for characterization of the sensor. (a) Photo of occlusion position test setup. (b) Output voltage of each ambient-light photodiode with the cover moving from left to right. (c) Photo of pressure test setup. (d) Output voltage of infrared photodiode under applied pressure. (e) Test results of the sensor with a finger.

contact position. This performance further facilitates the possibility of detecting the interaction sensory information, such as object detection, bending angles, and force measurements in a soft robotic gripper equipped with several M-SOWs on the interface between the gripper and objects, as illustrated in Fig. 1(e).

B. Fabrication

The fabrication of M-SOW involves the materials and dimensions of the dual-light-channel waveguide, as well as the electro-optical components, which consist of a single infrared Light-Emitting Diode (LED) and three photodiodes. Additionally, the fabrication process also includes their respective signal processing circuits. These technique details

are shown in Fig. 2. In Fig. 2 (a), it is illustrated that the fabrication of M-SOW proceeds in two steps: a) fabricating the flexible dual-light-channel waveguide (Silicone rubber 70A) by molding technologies; b) inserting the electro-optical components into both ends of waveguide (Infrared LED: IR91-21C, Infrared photodiode: PT17-21B, ambient-light photodiode: ALS-PT19-315C). The M-SOW prototype is shown in Fig. 2(b). Its length is 40mm, width is 10mm, thickness is 3.5mm. Subsequently, the proposed design is integrated into the circuits that perform the amplification and conversion of the photodiode's current into voltage signals as outputs. They also supply stable currents to the infrared LED, ensuring accurate current input, as depicted in Fig. 2(c).

III. EXPERIMENTAL VALIDATION

In this section, the ability to detect contact position and force along the M-SOW was verified. Two custom experimental platforms were constructed for testing the proposed design, as illustrated in Fig. 3(a) and Fig. 3(c). In Fig. 3(a), the M-SOW was fixed at a steel platform. A slim cover was suspended above the waveguide and connected to a slider, which was driven by a stepper motor. The displacements of the slider were recorded by a laser sensor (Panasonic, HG-C1100, $50\text{mm} \pm 35\text{mm}$). This configuration allowed the waveguide to independently validate its capacity for measuring contact position without any contact force applied to it. The experimental platform for testing the magnitude of contact force was shown in Fig. 3(c) where the waveguide was fixed at the platform. A force sensor (range: 0-20N) was mounted on the tip of a linear driven motor. In this way, when the motor moved linearly to press the waveguide, the force sensor recorded the magnitude of contact force.

A. Contact position measurement

Fig. 3(b) illustrates the relationship between the output voltages of the ambient light photodiodes at both ends of the channel. In this test, the stepper motor drove the slim cover from the left to the right side, and the output voltages of the waveguide were recorded as the cover progressed. The results show that both photodiodes have an initial maximum output voltage of approximately 0.92V when there is no covering. A noticeable voltage drop occurs when the slim cover reaches the top of the waveguide. As the cover continues to move to the right side, the voltage of the left photodiode increases while the right photodiode decreases, displaying a distinct voltage variation at different positions. This indicates that the photodiodes at the two ends of the channel can produce sets of voltage variations for detecting contact positions.

B. Contact force measurement

Fig. 3(d) demonstrates the relationship between the contact force and the output voltage of the infrared photodiode. Initially, the force sensor does not contact the waveguide, resulting in a maximum initial voltage of 0.29V. As the force sensor presses the waveguide, the output voltage decreases monotonically. This voltage drop is attributed to changes in the light path. When the contact pressure is applied to the waveguide, fewer infrared lights pass through the channel, leading to a reduction in the light power received by the photodiode. Consequently, the ability to measure contact force has been validated.

C. Mixed contact interaction demonstration

To verify the sensing performance of the proposed design under mixed contact situations, i.e., the combination of contact position and forces, we used one finger to make contact. The results are presented in Fig. 3(e). Initially, as the finger approaches the waveguide, the two ambient light photodiodes are capable of capturing this process. Subsequently, the finger carefully avoids covering the ambient light channel and presses the infrared channel. This demonstrates that the infrared photodiode can detect the applied pressure. Finally, when the finger simultaneously covers and presses the waveguide, voltage variations can be observed by all photodiodes. These three processes illustrate

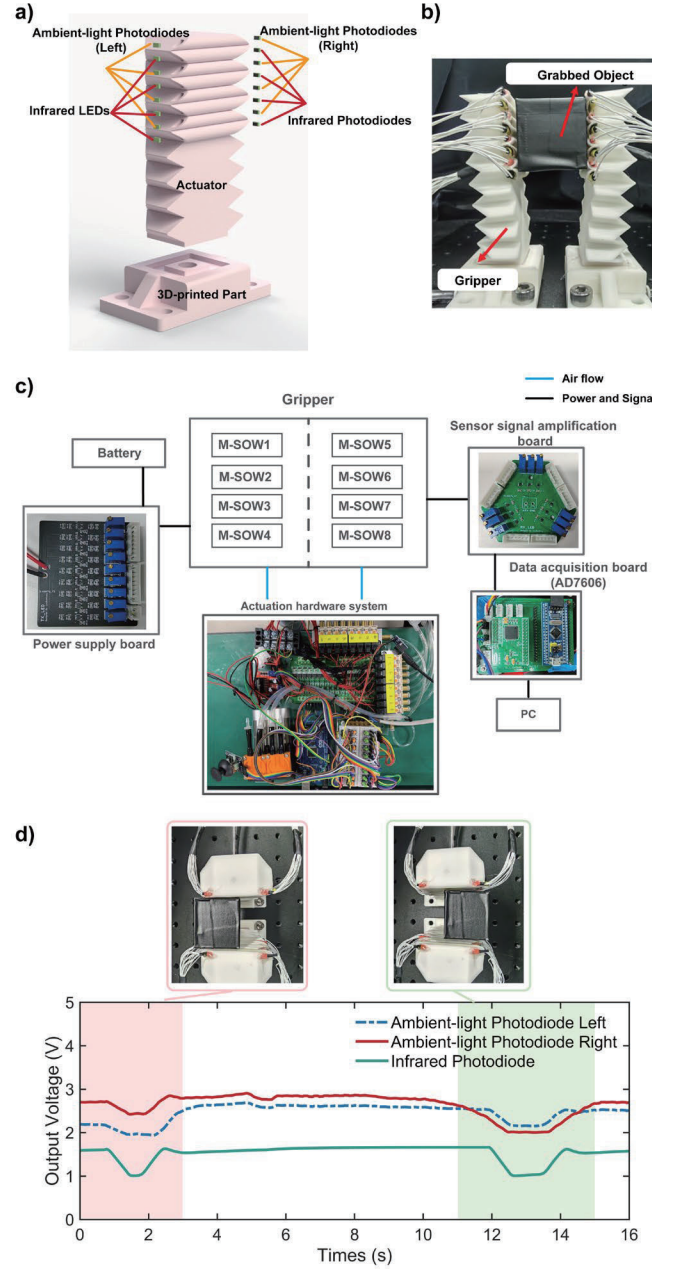


Fig. 4. Grasping experiment using a soft robot gripper equipped with m-sow. (a) Design of the soft robotic gripper. (b) Prototype of the soft robotic gripper. (c) Hardware system diagram of soft robotic gripper. (d) Output voltage of an M-SOW in a soft robot gripper when grasping objects at different positions

the contact sensing capabilities of the proposed approach.

IV. SOFT ROBOTIC DEMONSTRATION

The proposed M-SOW design has been integrated into a soft robotic gripper as a preliminary demonstration of its potential applications in soft robotics. The soft robotic gripper comprises a pair of bending soft actuators, each embedded with four parallel M-SOWs, as depicted in Fig. 4(a) and Fig. 4(b). Furthermore, an actuation system with pressure feedback control has been developed (Fig. 4(c)), consisting of

two independent channels that receive pneumatic pressure commands generated by the upper computer. This system employs pneumatic pressure sensors and on/off solenoid valves, enabling the underlying controller to accurately execute the desired pressure trajectory in real-time. With this configuration, a 3D-printed cubic object was tested to evaluate the sensing capabilities of the gripper. Three signal channels from three photodiodes in one M-SOW were plotted to demonstrate the grasped state of the object. The soft gripper was supplied with 80kPa air pressure. Initially, the object was grasped on the left side of the gripper, as shown in Fig. 4(d). The output voltage of the infrared photodiode decreased from 1.2V to around 0.5V, while the output voltages of the ambient light photodiodes also exhibited variation. Subsequently, the object's position was shifted to the right side. The infrared photodiode displayed almost the same voltage, but the groups of ambient light photodiodes showed differences. This demonstrates that the proposed M-SOW design enables the acquisition of interaction sensory information.

V. CONCLUSION

In this study, we presented the concept of M-SOW that utilizes multiple wavelengths of light as redundant sensory inputs for contact force and contact position measurements in soft robots. A M-SOW sensor design that combines external lights (wavelength from 380nm to 780nm) and internal infrared lights (wavelength from 780nm to 1100nm) was proposed. It was further fabricated to a TPU flexible dual-light-channel waveguide. One channel incorporated two ambient light photodiodes to receive external visible light, while the other integrated an infrared LED and an infrared photodiode to ensure internal light propagation. Consequently, the ambient light channel facilitated contact position measurements along the entire channel, while the infrared channel was employed to measure contact force magnitude. To validate these sensing capabilities, a series of experimental validation were conducted in two customized experimental platforms. The results demonstrated that the proposed design is capable of handling the detection of contact position and force during the interaction in the waveguide. Leveraging these characteristics, a soft robotic gripper was developed showing the possible soft robotic applications with capabilities of object interactions detection.

Future work will explore the methods of modeling and analyzing the multiple-channel signals. Compensation mechanisms for sensors under unstable lighting conditions and the merits of the sensors to the robotic control in more sophisticated tasks will also be further investigated.

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